Marwadi University logo	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:** Compute and analyze the correlation between three audio tracks: the original song, the karaoke (music-only) version, and a completely different song.	
Open-ended : 3	**Date: 02/08/2026**	**Enrollment No: 92400133037**

Aim: Compute and analyze the correlation between three
audio tracks:
1. the original song,
2. the karaoke (music-only) version, and
3. a completely different song.

Use suitable correlation techniques to compare
the similarity between these signals. Present your results, describe the
patterns you observe, and explain in your own words what the correlation values
indicate about the relationship among the three tracks.

Program/Code:

```
pip install librosa scipy matplotlib numpy

# The librosa.load() function loads the audio files into NumPy arrays.

# The parameter sr=22050 resamples all audio files to 22050 Hz, ensuring that all signals

#have the same sampling rate.

# The parameter mono=True converts stereo audio into a single channel. This simplifies the
#comparison because each audio file is represented by one waveform instead of two.

import librosa

original, sr = librosa.load(
    "original.mp3",
    sr=22050,
    mono=True
)


karaoke, _ = librosa.load(
    "karaoke.mp3",
    sr=22050,
    mono=True
)
```

```
different, _ = librosa.load(
    "different.mp3",
    sr=22050,
    mono=True
)


# Only the first 30 seconds of each audio file are used.
#The total number of samples is calculated as
#samples = sampling rate × duration
# 22050 × 30 = 661500 samples
# Using equal-duration clips reduces computation time and ensures a fair comparison.


duration = 30
samples = sr * duration


original = original[:samples]
karaoke = karaoke[:samples]
different = different[:samples]


# Although the songs are trimmed to 30 seconds, minor differences in length may still exist
#due to encoding or metadata. The smallest signal length is determined using min() and all
#signals are trimmed to this common length. This ensures that correlation is computed
#correctly


min_len = min(len(original), len(karaoke), len(different))
original = original[:min_len]
karaoke = karaoke[:min_len]
```

```
different = different[:min_len]


# Different songs may have different recording volumes. Normalization scales every signal so
#that its maximum amplitude becomes 1.

#This removes the effect of loudness and ensures that the comparison depends only on the
#waveform characteristics rather than recording volume.


import numpy as np

original = original / np.max(np.abs(original))

karaoke = karaoke / np.max(np.abs(karaoke))

different = different / np.max(np.abs(different))


# The function np.corrcoef() computes the Pearson Correlation Coefficient, which
#measures the linear relationship between two signals.


corr_ok = np.corrcoef(original, karaoke)[0, 1]

corr_od = np.corrcoef(original, different)[0, 1]

corr_kd = np.corrcoef(karaoke, different)[0, 1]


print("Original vs Karaoke :", corr_ok)

print("Original vs Different:", corr_od)

print("Karaoke vs Different :", corr_kd)


# The correlate() function computes the cross-correlation between two signals. Unlike
#Pearson correlation, cross-correlation shifts one signal relative to the other and computes
#similarity at each possible lag.

#mode='full' ensures that correlation is calculated for every possible alignment between the
#two signals.
```

```
from scipy.signal import correlate
cross_ok = correlate(original, karaoke, mode="full")
cross_od = correlate(original, different, mode="full")
cross_kd = correlate(karaoke, different, mode="full")


#plotting
# Each graph represents the variation in similarity as one signal is shifted relative to the other.
#The central region of the graph corresponds to the best alignment between the signals.


import matplotlib.pyplot as plt
plt.figure(figsize=(18,5))
plt.subplot(1,3,1)
plt.plot(cross_ok)
plt.title("Original vs Karaoke")


plt.subplot(1,3,2)
plt.plot(cross_od)
plt.title("Original vs Different")


plt.subplot(1,3,3)
plt.plot(cross_kd)
plt.title("Karaoke vs Different")


plt.tight_layout()
plt.show()
```

```
print(np.max(cross_ok))

print(np.max(cross_od))

print(np.max(cross_kd))



from scipy.signal import correlate

import numpy as np

cross_ok = correlate(original, karaoke, mode='full')

cross_od = correlate(original, different, mode='full')

cross_kd = correlate(karaoke, different, mode='full')


# Normalize by signal energies

# Raw cross-correlation values depend on signal length and amplitude. To enable fair
comparison, the cross-correlation is normalized using the energy of both signals. This
converts the values into normalized similarity scores


cross_ok = cross_ok / np.sqrt(np.sum(original**2) * np.sum(karaoke**2))

cross_od = cross_od / np.sqrt(np.sum(original**2) * np.sum(different**2))

cross_kd = cross_kd / np.sqrt(np.sum(karaoke**2) * np.sum(different**2))


#  np.max(cross_ok)---The maximum value of the normalized cross-correlation represents the
#highest similarity between the two signals after considering all possible shifts.

#This value is used as the final similarity measure.


print("Max Original vs Karaoke :", np.max(cross_ok))

print("Max Original vs Different :", np.max(cross_od))

print("Max Karaoke vs Different :", np.max(cross_kd))
```

Results:

Original vs Karaoke : 0.01816849955644602

Original vs Different: -0.0012579030610307864

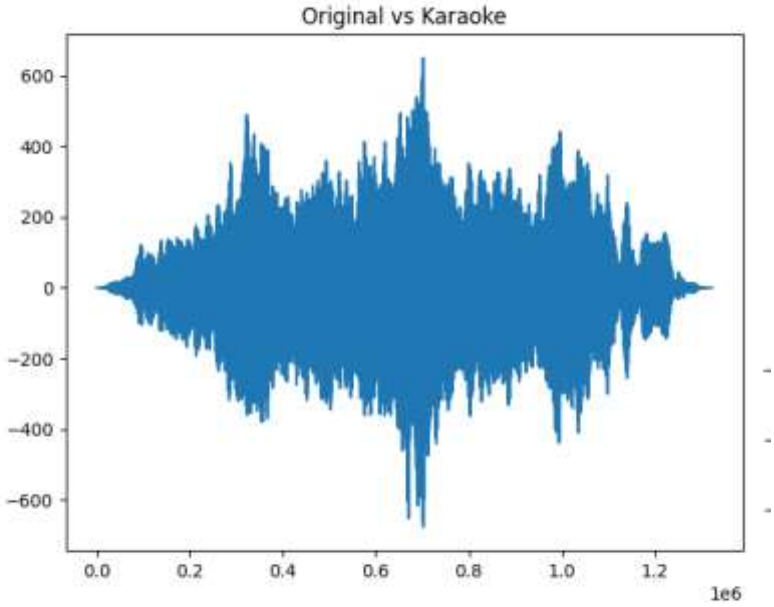
Karaoke vs Different : 0.0007607061375107056

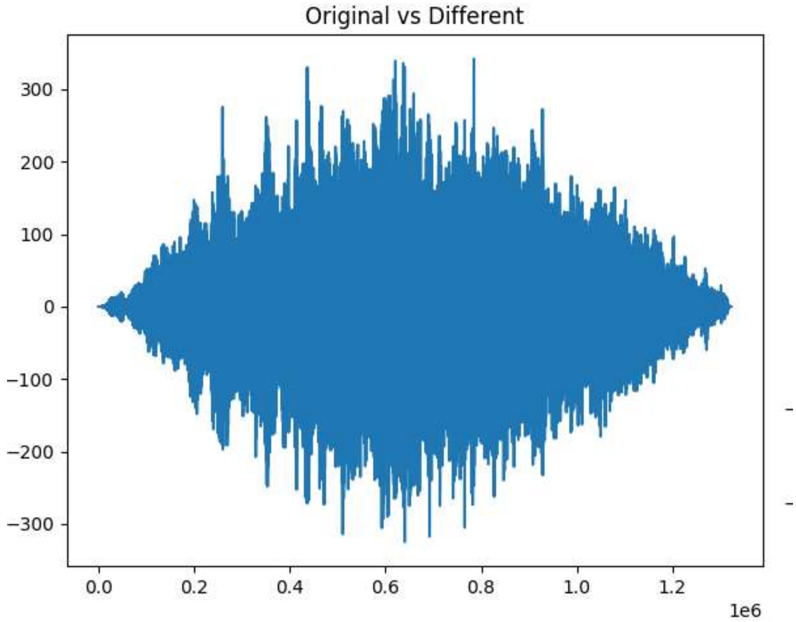
Max Original vs Karaoke : 0.04984906

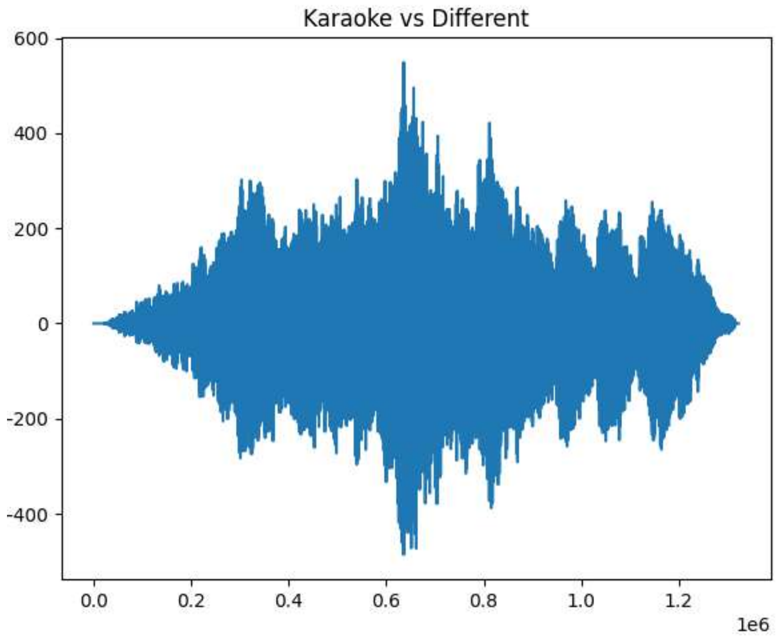
Max Original vs Different : 0.016466353

Max Karaoke vs Different : 0.02751717

Comparison	**Pearson**	**Max Cross Correlation**
Original vs Karaoke	0.0182	0.0498
Original vs Different	-0.0013	0.0165
Karaoke vs Different	0.0008	0.0275







Analysis:

The Pearson Correlation Coefficient between the original song and the karaoke version was 0.0182, while the Maximum Normalized Cross-Correlation was 0.0498. Although these values are relatively small, they are the highest among all three comparisons. This indicates that the original song and its karaoke version share the greatest similarity because they originate from the same musical composition and contain the same melody and instrumental arrangement. The Pearson correlation remains low because it compares the signals sample by sample, and the removal of vocals in the karaoke version significantly changes the waveform.

The comparison between the original song and the different song produced a Pearson correlation of −0.0013 and a Maximum Normalized Cross-Correlation of 0.0165. Both values are very close to zero, indicating that the two audio signals have almost no linear relationship or waveform similarity. This is expected because the songs have different melodies, rhythms, tempos, and musical structures.

Similarly, the karaoke version and the different song produced a Pearson correlation of 0.0008 and a Maximum Normalized Cross-Correlation of 0.0275. These values are also very low, confirming that the karaoke track and the unrelated song are largely dissimilar. The slightly higher cross-correlation compared to the Original–Different pair is minor and may be due to common instrumental components or recording characteristics rather than any meaningful musical similarity.

Overall, the results demonstrate that Maximum Normalized Cross-Correlation is more effective than Pearson Correlation for comparing audio signals. Pearson correlation performs a strict sample-by-sample comparison and is highly sensitive to changes such as missing vocals, timing differences, and waveform variations. In contrast, cross-correlation compares the signals over different time shifts and identifies the best possible alignment, making it a more suitable technique for measuring similarity between audio recordings.

Experiment1: Effect of Audio Duration

To study how the length of the audio clip affects Pearson correlation and normalized cross-correlation.

Durations to test

We'll use:

- 10 seconds
- 20 seconds
- 30 seconds
- 45 seconds
- 60 seconds

```
import librosa

import numpy as np

from scipy.signal import correlate


durations = [10, 20, 30, 45, 60]


print("-" * 90)

print(f"{'Duration':<10}{'Pearson OK':<15}{'Pearson OD':<15}{'Pearson KD':<15}"

      f"{'Cross OK':<15}{'Cross OD':<15}{'Cross KD':<15}")

print("-" * 90)


for duration in durations:


    # Reload the audio each time

    original, sr = librosa.load("original.mp3",

                                sr=22050,

                                mono=True)


    karaoke, _ = librosa.load("karaoke.mp3",

                              sr=22050,

                              mono=True)


    different, _ = librosa.load("different.mp3",

                                sr=22050,

                                mono=True)
```

```
samples = duration * sr

    original = original[:samples]
    karaoke = karaoke[:samples]
    different = different[:samples]

    min_len = min(len(original), len(karaoke), len(different))

    original = original[:min_len]
    karaoke = karaoke[:min_len]
    different = different[:min_len]

    # Normalize
    original = original / np.max(np.abs(original))
    karaoke = karaoke / np.max(np.abs(karaoke))
    different = different / np.max(np.abs(different))

    # Pearson
    p_ok = np.corrcoef(original, karaoke)[0,1]
    p_od = np.corrcoef(original, different)[0,1]
    p_kd = np.corrcoef(karaoke, different)[0,1]

    # Cross Correlation
    c_ok = correlate(original, karaoke, mode='full')
    c_od = correlate(original, different, mode='full')
    c_kd = correlate(karaoke, different, mode='full')
```

```
c_ok = c_ok / np.sqrt(np.sum(original**2) * np.sum(karaoke**2))
c_od = c_od / np.sqrt(np.sum(original**2) * np.sum(different**2))
c_kd = c_kd / np.sqrt(np.sum(karaoke**2) * np.sum(different**2))


print(f"{duration:<10}"
    f"{p_ok:<15.6f}"
    f"{p_od:<15.6f}"
    f"{p_kd:<15.6f}"
    f"{np.max(c_ok):<15.6f}"
    f"{np.max(c_od):<15.6f}"
    f"{np.max(c_kd):<15.6f}")
```

Duration (s)	Pearson (Original–Karaoke)	Pearson (Original–Different)	Pearson (Karaoke–Different)	Max Cross-Correlation (Original–Karaoke)	Max Cross-Correlation (Original–Different)	Max Cross-Correlation (Karaoke–Different)
10	-0.054041	-0.001313	0.011337	0.147417	0.020863	0.038832
20	-0.000199	0.001989	0.001441	0.063784	0.021477	0.023646
30	0.018168	-0.001258	0.000761	0.049849	0.016466	0.027517
45	0.011891	-0.001036	-0.001295	0.079920	0.016318	0.020935
60	0.000758	-0.002600	-0.001981	0.062730	0.014820	0.016309

All Pearson coefficients are very close to **zero**.

Pearson correlation compares the signals **sample by sample**.

Even though the original and karaoke versions represent the same song, the karaoke track has the vocals removed.

Mathematically,

$$\text{Original} = \text{Music} + \text{Vocals}$$

whereas

$$\text{Karaoke} = \text{Music}$$

Because thousands of waveform samples differ due to the missing vocals, the sample-wise similarity becomes very small.

Therefore, increasing the duration does not necessarily increase the Pearson correlation. Instead, the average similarity over a longer period may even decrease as more differing waveform sections are included.

Unlike Pearson correlation, the cross-correlation values are consistently higher for the Original–Karaoke pair than for the other two comparisons.

This indicates that cross-correlation successfully detects the greatest similarity between the related tracks, even though the exact waveform samples differ.

The first 10 seconds of "Perfect" mainly contain:

- piano,
- guitar,
- soft background music,

and relatively fewer vocal variations.

Therefore, the original and karaoke versions are naturally more alike during this segment.

As the duration increases, more vocal content enters the original song. Since the karaoke version lacks vocals, the similarity decreases.

This explains why:

$$0.147 > 0.064 > 0.050$$

The 45-second interval likely includes an instrumental section where the original and karaoke become more similar again.

Hence the cross-correlation increases.

Between Original and the Different Song the normalized cross-correlation remains approximately 0.02 for all the durations.

This indicates that the two songs are unrelated.

Changing the duration does not significantly affect the similarity because their melodies, rhythms, and waveform structures are fundamentally different.

Between Karaoke and the Different Song the values remain between 0.016–0.039

These are also very low, confirming that the karaoke version shares very little similarity with the unrelated song.

Graph	**What it tells us**	**Interpretation**
Cross-Correlation	Measures similarity after allowing time shifts	Better for audio comparison
Pearson Correlation	Measures sample-by-sample linear similarity	Highly sensitive to waveform differences

Experiment 2: Effect of Sampling Rate on Audio Correlation

To investigate how changing the sampling rate affects the Pearson Correlation Coefficient and the Maximum Normalized Cross-Correlation between the original song, its karaoke version, and a different song.

```
import librosa

import numpy as np

from scipy.signal import correlate


sampling_rates = [8000, 16000, 22050, 44100]


duration = 30


print("-"*95)

print(f"{'Sampling Rate':<15}"

      f"{'Pearson OK':<15}"
```

```
        f"{'Pearson OD':<15}"
        f"{'Pearson KD':<15}"
        f"{'Cross OK':<15}"
        f"{'Cross OD':<15}"
        f"{'Cross KD':<15}")
    print("-"*95)

    for sr in sampling_rates:

        original, _ = librosa.load(
            "original.mp3",
            sr=sr,
            mono=True
        )

        karaoke, _ = librosa.load(
            "karaoke.mp3",
            sr=sr,
            mono=True
        )

        different, _ = librosa.load(
            "different.mp3",
            sr=sr,
            mono=True
        )
```

```
samples = duration * sr

original = original[:samples]
karaoke = karaoke[:samples]
different = different[:samples]

min_len = min(len(original), len(karaoke), len(different))

original = original[:min_len]
karaoke = karaoke[:min_len]
different = different[:min_len]

# Normalize
original = original / np.max(np.abs(original))
karaoke = karaoke / np.max(np.abs(karaoke))
different = different / np.max(np.abs(different))

# Pearson
p_ok = np.corrcoef(original, karaoke)[0,1]
p_od = np.corrcoef(original, different)[0,1]
p_kd = np.corrcoef(karaoke, different)[0,1]

# Cross Correlation
c_ok = correlate(original, karaoke, mode='full')
c_od = correlate(original, different, mode='full')
c_kd = correlate(karaoke, different, mode='full')
```

```
c_ok = c_ok / np.sqrt(np.sum(original**2) * np.sum(karaoke**2))
c_od = c_od / np.sqrt(np.sum(original**2) * np.sum(different**2))
c_kd = c_kd / np.sqrt(np.sum(karaoke**2) * np.sum(different**2))


print(f"{sr:<15}"
    f"{p_ok:<15.6f}"
    f"{p_od:<15.6f}"
    f"{p_kd:<15.6f}"
    f"{np.max(c_ok):<15.6f}"
    f"{np.max(c_od):<15.6f}"
    f"{np.max(c_kd):<15.6f}")
```

The experimental results show that changing the sampling rate had very little influence on the computed correlation values. The Pearson correlation coefficient for the Original–Karaoke pair remained nearly constant, changing only from 0.018640 at 8000 Hz to 0.018151 at 44100 Hz. Similarly, the Maximum Normalized Cross-Correlation changed only slightly from 0.051028 to 0.049815. The differences are very small, indicating that the similarity measurement is almost independent of the selected sampling rate within the tested range.

The Original–Karaoke pair consistently produced the highest correlation values among all three comparisons. This is expected because both tracks originate from the same musical composition and share the same melody, rhythm, and instrumental arrangement. Although the karaoke version does not contain vocals, it still retains enough common musical information to produce a higher similarity than the unrelated song.

The Original–Different and Karaoke–Different comparisons remained almost constant throughout the experiment, with cross-correlation values close to 0.016–0.028 and Pearson correlation values close to zero. These low values indicate that the different song has a completely different waveform, melody, rhythm, and musical structure. Changing the sampling rate cannot increase the similarity between two fundamentally different audio signals.

Therefore, the sampling rate primarily controls the number of samples recorded per second, thereby affecting the resolution of the digital representation of the signal. However, resampling does not alter the underlying musical content of the audio. Since all the tested sampling rates were sufficiently high to preserve the important characteristics of the songs, the correlation values remained almost unchanged. This demonstrates that the similarity between audio signals is determined mainly by the actual signal content rather than the sampling resolution.

Conclusion:

In this work, the similarity between an original song, its karaoke version, and a completely different song was analyzed using Pearson Correlation and Maximum Normalized Cross-Correlation. The results showed that the original song and its karaoke version consistently exhibited the highest similarity, while comparisons involving the unrelated song produced significantly lower correlation values.

Pearson Correlation produced values close to zero because it compares audio signals on a sample-by-sample basis and is highly sensitive to waveform variations caused by missing vocals and timing differences. Maximum Normalized Cross-Correlation provided a more meaningful measure of similarity by considering different time shifts between the signals.

The parameter analysis demonstrated that the duration of the selected audio segment influences the measured similarity more than the sampling rate. Instrumental sections generally produced higher similarity than sections dominated by vocals, while varying the sampling rate between 8000 Hz and 44100 Hz had only a negligible effect on the correlation values.

Overall, the experiment confirms that cross-correlation is a more suitable technique than Pearson correlation for comparing audio signals and identifying similarities between related musical recordings.